

Erwin Quanten

Staff Software Engineer – React, Node.js, TypeScript, AWS

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PROFESSIONAL SUMMARY

Staff Software Engineer with strong experience building enterprise **React 16–18, Node.js 12–20, TypeScript 3.9–5.x, Express 4.x, NestJS 7–10**, and **AWS** platforms across SaaS, logistics, healthcare, fintech, and industrial software industries. Specialized in modernizing legacy applications, designing scalable APIs, improving frontend performance, implementing secure authentication, resolving production defects, and migrating older systems into cloud-native architectures while improving delivery speed, reliability, and maintainability.

SKILLS

- **Frontend:** React 16–18, Next.js 10–14, TypeScript 3.9–5.x, JavaScript ES6+, Redux Toolkit, React Query, Zustand, HTML5, CSS3, SCSS, Tailwind CSS, Material UI
- **Backend:** Node.js 12–20, Express 4.x, NestJS 7–10, REST API, GraphQL, WebSocket, Microservices, JWT, OAuth2, RBAC
- **Cloud & DevOps:** AWS EC2, Lambda, ECS, S3, CloudFront, RDS, Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Jenkins
- **Database:** PostgreSQL 10–15, MySQL 5.7–8, MongoDB 4–6, Redis 5–7, Elasticsearch 7–8
- **Testing & Quality:** Jest, React Testing Library, Cypress, Playwright, Supertest, SonarQube, ESLint, Prettier
- **Architecture:** Monorepo, Domain-Driven Design, Event-Driven Architecture, API Gateway, CI/CD, Observability, Performance Optimization

EXPERIENCE

Lightedge — Lead Software Engineer

Remote/ Full time

May 2022 – Apr 2026

Houston, TX

- Led development of a secure **React 18, Node.js 18, TypeScript 5.x**, and **AWS** platform for healthcare operations, improving dashboard response speed by **32%** through code splitting, memoization, and API response caching.
- Designed scalable **NestJS 9** backend services with modular architecture, request validation, guards, interceptors, and structured logging to reduce production incident investigation time by **27%**.
- Migrated legacy **Express 4.x** services into **NestJS 9**, keeping backward-compatible REST contracts while improving maintainability and reducing duplicated business logic by **24%**.
- Implemented secure authentication using **OAuth2, JWT**, refresh tokens, and role-based access control, solving inconsistent permission bugs that caused users to see incorrect workflow actions.
- Built reusable **React 18** component libraries with TypeScript generics, form validation, and accessibility patterns, reducing repeated UI implementation effort by **29%**.
- Resolved a critical production memory leak in a Node.js worker by analyzing heap snapshots, fixing unclosed database connections, and adding runtime monitoring alerts.
- Improved CI/CD pipelines using **GitHub Actions**, Docker image caching, Jest coverage gates, and automated deployment checks, reducing release time by **35%**.
- Refactored slow PostgreSQL queries with indexing, pagination, and query-plan analysis, reducing high-volume report generation time by **41%**.
- Implemented real-time status updates with **WebSocket** and Redis pub/sub, improving operational visibility for support teams and reducing manual refresh issues.
- Introduced structured error handling with custom exception filters and correlation IDs, making frontend and backend failures easier to trace across distributed services.
- Created automated tests with **Jest, React Testing Library, Cypress**, and **Supertest**, increasing regression coverage and preventing recurring checkout and authorization bugs.
- Mentored engineers on **React performance, Node.js architecture**, and TypeScript patterns, improving pull request quality and reducing repeated review comments across the team.

Syscom Digital — Staff Software Engineer

Remote/ Full time

Nov 2017 – Apr 2022

Bucharest, Romania

- Architected a financial operations platform using **React 17, Node.js 16, TypeScript 4.x**, and **PostgreSQL 13**, improving transaction review workflow completion speed by **28%**.

- Built backend APIs with **Express 4.x** and later migrated key modules to **NestJS 8**, improving dependency organization and reducing service onboarding complexity.
- Implemented complex frontend state management using **Redux Toolkit** and **React Query**, solving stale data issues in multi-step financial approval screens.
- Migrated class-based React components into functional components with hooks, improving readability and reducing frontend defect density by **22%**.
- Solved a production issue where duplicate payment events were processed by adding idempotency keys, database constraints, and retry-safe queue consumers.
- Integrated **Redis 6** caching for frequently requested account and permission data, reducing backend load by **31%** during business-hour traffic spikes.
- Created reusable API client utilities with typed responses, centralized error handling, and token refresh logic to reduce inconsistent frontend integration patterns.
- Improved Docker-based local development by standardizing environment variables, seed scripts, and service startup order, reducing setup time for new developers by **38%**.
- Implemented audit logging for sensitive financial actions, helping compliance teams trace user activity without slowing normal application workflows.
- Collaborated with product owners and QA teams to convert unclear business rules into tested workflows, reducing rework during release validation.

Tessenderlo Group — Senior Software Engineer

Remote/ Full time

Aug 2015 – Oct 2017

Brussel Metropolitan Area

- Developed internal supply-chain applications using **React 16**, **Node.js 12–14**, **Express 4.x**, and **SQL Server**, improving operational planning visibility by **26%**.
- Modernized legacy jQuery screens into reusable **React 16** modules, reducing page load time by **34%** and improving maintainability for plant operations users.
- Built REST APIs for inventory, shipment, and production data, adding validation layers that reduced incorrect manual entries by **23%**.
- Solved recurring timezone and date-format bugs in logistics workflows by centralizing date handling, API serialization, and frontend display rules.
- Implemented background processing for large import files using Node.js streams, reducing failed uploads and improving processing speed by **37%**.
- Created role-based UI behavior for planners, supervisors, and administrators, preventing unauthorized workflow changes and reducing support tickets.
- Migrated deployment scripts from manual server updates into repeatable CI/CD steps, improving release consistency and reducing deployment rollback cases.
- Optimized SQL queries and API pagination for high-volume production records, reducing timeout errors during monthly reporting cycles.
- Worked closely with plant stakeholders to translate operational pain points into practical features that improved daily planning and exception handling.

Tessenderlo Group — Software Engineer

Hybrid/ Full time

Sep 2011 – Jul 2015

Brussel Metropolitan Area

- Built business applications with **JavaScript ES6**, early **React 15–16**, **Node.js 8–10**, and REST services to support procurement and production teams.
- Introduced component-based frontend development gradually, replacing fragile page-level scripts with reusable UI sections and clearer state handling.
- Implemented backend services with **Express 4.x**, improving integration between internal tools, ERP exports, and reporting databases.
- Resolved recurring form validation issues by creating shared validation utilities, improving data accuracy and reducing manual correction effort by **21%**.
- Improved legacy application performance by removing duplicated API calls, optimizing payload size, and caching stable reference data on the client.
- Supported migration from older JavaScript patterns into TypeScript-ready code structure, making future upgrades safer and easier for the team.
- Created deployment documentation and environment setup guides, reducing onboarding issues and helping developers reproduce production-like behavior locally.
- Collaborated with QA to identify regression-prone workflows and added automated test coverage around critical order and approval paths.

Tessenderlo Group — Software Developer*Hybrid/ Full time***Jun 2009 – Aug 2011***Brussel Metropolitan Area*

- Developed internal web modules using widely adopted **JavaScript**, **jQuery 1.x**, **HTML**, **CSS**, and server-rendered application patterns used during that period.
- Maintained legacy business workflows and fixed browser compatibility issues that affected production users across procurement and reporting teams.
- Improved slow reporting pages by optimizing SQL queries, reducing unnecessary joins, and improving server-side filtering for large datasets.
- Supported gradual modernization by separating presentation logic from business logic, preparing several modules for later React and API-based migration.
- Resolved recurring production bugs by analyzing logs, reproducing user steps, and documenting root causes to prevent the same issue from returning.

EDUCATION

KU Leuven — Master's Degree, Computer Science**Sep 2003 – May 2009**